Foundations of Generative AI (ICT4403) Assignment 2 Submission Document

Autoregressive Text Generation System using Simple RNN, Bidirectional RNN, LSTM, and GRU

1. Submission Details

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Assignment Title	Foundations of Generative AI (ICT4403) - Assignment 2
Title of Developed System	Comparative Autoregressive Text Generation using Simple RNN, Bidirectional RNN, LSTM, and GRU Architectures
Dataset Name and Link	The Adventures of Sherlock Holmes by Arthur Conan Doyle https://www.gutenberg.org/ebooks/1661
GitHub Repository Link	https://github.com/Kshiku12/Autoregressive-Text-Generation-RNN-LSTM-GRU

2. Executive Summary & Experimental Results

This project develops and empirically evaluates an autoregressive next-word language modeling system across four fundamental recurrent neural architectures: Simple RNN, Bidirectional RNN, LSTM, and GRU. All four models were trained under strictly controlled, identical experimental conditions: same dataset split (85% train, 15% validation), vocabulary size (4,406 words, 97.14% token coverage), embedding dimension (128), hidden units (256), batch size (128), context window (20 words), learning rate (0.002 with Adam), and cross-entropy loss.

3. Key Findings & Best-Performing Model

- **Quantitative Superiority:** LSTM and GRU decisively outperform Simple RNN and Bidirectional RNN in validation loss and perplexity. LSTM achieved the lowest validation loss (4.8432) and lowest perplexity (126.88), with top-1 next-word accuracy of 18.79% and top-5 accuracy of 41.44%.
- **Architectural Efficiency:** GRU closely matches LSTM performance (Val Loss 4.8807, PPL 131.73, Top-1 Acc 18.12%) while requiring fewer parameters (1,992,758 vs. 2,091,574) and faster training time per epoch.

- **Temperature Influence:** At T = 0.2, models degenerate into repetitive loops. At T = 1.2, models produce high lexical diversity but lose grammatical coherence. Temperature T = 0.7 offers the optimal trade-off between coherence and diversity.
- **Final Selection:** LSTM is selected as the most effective model for autoregressive generation on this corpus.